

ARIF WINANDA LUBIS

Software & Hardware Development and AI Engineer Enthusiast

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Summary

Information Technology Graduate from Telkom University and aspiring Full Stack Developer with solid foundation in end-to-end web and mobile application development, database development, and API Integration. Proven track record in student leadership and functional collaboration, with strong analytical skills (problem-solving), active communication, and organizational capabilities. Dedicated to do work to make a new capabilities of learn and continuous innovation to deliver scalable with optimality with high performance digital solutions in dynamic team environments.

Education

Telkom University

Bandung

Bachelor of Information Technology

2026

- GPA: 3.40 / 4.00
- Coursework: Object-Oriented Programming, Database Management Systems, Web & Mobile Application Development, Software Engineering, Computer Networks & Security, Artificial Intelligence & Machine Learning.

Experience

Lecturer Assistant of IoT Intelligent Systems

Bandung

Faculty of Informatics Telkom University

February 2026 – June 2026

- Assisted senior lecturers in delivering core concepts of Artificial Intelligence, Machine Learning, and Internet of Things (IoT) architectures to students.
- Guided and mentored students through the data preprocessing, cleaning, and optimization phases to ensure highquality data management for their IoT system projects.
- Evaluated student practical assignments, lab protocols, and embedded hardware configurations utilizing microcontrollers and communication protocols.
- Provided technical troubleshooting and advisory support for student-designed smart system integration, maximizing data collection reliability and project outcomes.

Backend Developer

Bandung

Web Project By Sekretariat Daerah Pemerintah Provinsi Jawa Barat

September 2025 – January 2026

- Developed a comprehensive, full-scale internal employee enterprise platform incorporating a daily work reporting module and an asset inventory tracker.
- Architected and deployed robust RESTful APIs built on top of the Laravel (PHP) framework to handle seamless back-end transactions.
- Implemented secure Role-Based Access Control (RBAC) layers to protect sensitive government data assets and streamline internal cross-departmental tracking.

Web Developer Internship

Bandung

Dinas Komunikasi dan Informatika Provinsi Jawa Barat

July – September 2025

- Engineered and developed the ITSO internal web application utilizing the Laravel (PHP) framework as the primary core architecture.
- Built dynamic filtering capabilities handling complex multi-variable data tables using JavaScript DataTables and designed interactive data visualization charts with Chart.js.
- Managed secure web archival systems by integrating Role-Based Access Control (RBAC) to ensure strict data governance.
- Optimized automated background server management and enhanced data accessibility by configuring automated Linux cron-jobs.

Project Experience

LILY'S CATERING: Enterprise B2B Catering & Financial Management Platform

Tech Stack: NextJS, Node.JS (Express), MongoDB, Rest API, Git, Midtrans Payment Gateway.

<https://lily-catering-seven.vercel.app/>

- Engineered a robust B2B corporate catering and financial management platform using Next.js and Node.js, featuring multi-vendor catalog management, automated scheduling for institutional orders, and secure digital payment processing.
- Developed scalable backend architecture and RESTful APIs with Node.js and MongoDB (Mongoose), implementing dynamic data aggregation pipelines to calculate gross merchandise value (GMV), payment gateway fees, and net profit margins in real-time.
- Integrated Midtrans payment gateway with automated webhooks and transaction state-syncing, paired with an automated expense tracker (COGS) to monitor operational costs including raw kitchen ingredients, logistics, and utilities.
- Designed an interactive vendor dashboard with financial analytics, digital invoice generators, and a structured payout request system following clean code standards, JWT authentication, and rigorous version control.

SANDITEL: Asset Management & Employee Reporting System

Tech Stack: ReactJS, PHP Laravel, PostgreSQL, Rest API, Git.

<https://fe-sanditel-blush.vercel.app/>

- Engineered a comprehensive internal management web application for the Telecommunications Section of the West Java Provincial Secretariat, featuring inventory asset tracking and daily employee reporting modules.
- Optimized core web system functionalities, including application logic improvements, seamless REST API integration between ReactJS and Laravel, and structured relational database queries.
- Applied Clean Code standards, rigorous feature debugging, and version control to deliver secure, highly maintainable, and reliable digital solutions for bureaucratic administration.

Melanoma Skin Cancer Detection & Explainable AI

Tech Stack: Python, PyTorch, ReXNet-150, Grad-CAM, Data Processing, Git.

[arifwinanda22/project_MelanomaDetection](https://github.com/arifwinanda22/project_MelanomaDetection)

- Engineered a Deep Learning classification pipeline utilizing the ReXNet-150 architecture to accurately detect melanoma skin cancer from medical imagery.
- Integrated Explainable AI (Grad-CAM) techniques to generate visual feature heatmaps, enhancing model interpretability and diagnostic transparency for computer aided diagnosis (CAD) systems.
- Authored and successfully published the comprehensive research methodology and experimental benchmarks in an IEEE-indexed international conference proceeding.

Smart Irrigation System & Educational IoT Modul

Tech Stack: C++, ESP32, MQTT Protocol, IoT Sensors, Mobile Applications, Sensor & Actuators.

https://arifwinanda22.github.io/Arif_Portofolio/

- Engineered an automated smart irrigation prototype using ESP32 microcontrollers to enable real-time environmental monitoring (soil moisture, temperature) and automated water pump actuation.
- Integrated the lightweight MQTT protocol for seamless sensor-to-cloud data streaming and remote monitoring/control via a dedicated mobile dashboard.
- Deployed the system as a hands-on educational demonstration and teaching module for IoT technology workshops aimed at school teachers and students.

SmartComp Multi-Platform Web Service & Mobile Ecosystem

Tech Stack: Laravel, React, Flutter, Dart, MySQL, Firebase Firestore, Restful API, & GIT

- Designed and developed an integrated multi-platform ecosystem connecting web interfaces (React & Laravel) and mobile applications (Flutter) for competition and workshop discovery.
- Built scalable RESTful APIs in Laravel and managed data pipelines across relational (MySQL) and real-time cloud databases (Firebase Firestore) for user authentication, skill management tracking, and dynamic event organizer dashboards
- Applied Clean Code principles, structured feature testing, and Git version control to ensure seamless end-to-end performance, responsive UI, and system security across web and mobile clients.

Organizational Experience

Student Association of Information Technology Student (HMIT) Telkom University

Member and Chairman

Bandung

2024 – ~~2025~~

- Served as the Chief Executive Officer to lead, direct, and oversee strategic operations, financial budgeting, and work programs for over 58 (fifty eight) active committee members.
- Spearheaded structural organizational reforms and strategic planning that increased student engagement and maximized collaborative partnerships with external institutions.
- Supervised the execution of large-scale student events and cross-functional working plans, ensuring all milestones were achieved within deadlines and under budgetary controls.

- Acted as the primary representative to negotiate, align goals, and maintain relations with university stakeholders, faculty members, and external industry partners.

Society of Renewable Energy (SRE) – Telkom University Member

Bandung
2023- 2024

- Actively participated in focused study groups and workshops exploring new developments in renewable energy technologies, sustainability frameworks, and green digital solutions.
- Assisted in organizing student-led seminars and community awareness campaigns to promote energy conservation practices and sustainability within the university environment.

Certification & Training

Artificial Intelligence use from Azure AI

Feb 2026 – Apr 2026

Elevaite AI by Microsoft

- Completed the preparation course for Azure AI Fundamentals (AI-900) organized by Elevaite AI in collaboration with Microsoft and KOMDIGI. Mastered core Artificial Intelligence and Machine Learning concepts, including the implementation of Microsoft Azure AI services guided by Responsible AI principles.

Java Programming

Sep 2024 – Jan 2025

Udemy

- Finished a 64.5-hour intensive course covering Java Fundamentals, Object-Oriented Programming (OOP), exception handling, and Standard Classes.

Web Development

Aug 2024 – Jan 2025

CODING STUDIO

- Completed comprehensive training encompassing Front-End Web Development fundamentals and advanced dynamic website architecture using the Laravel 9 framework.
- Acquired end-to-end competencies in designing responsive interfaces and managing database-driven architectures

Publications & Achievement

International Conference on Information and Communication Technology and Systems (ICICyTA)

2025

- Co-authored and successfully published an international academic research paper accepted into the ICICyTA 2025 conference proceedings.
- Contribute in-depth research on integrating Machine Learning models, specifically utilizing ReXNet-150 an Convolutional Neural Network (CNN) architectures for image based skin classification tasks.

Skills

- **Programming Languages & Frameworks:** PHP (Laravel), JavaScript, Java, NextJS, RESTful APIs, Android Mobile App Development.
- **Web & Front-end Tools:** HTML, CSS, JavaScript DataTables, Chart.js, Interactive Data Visualization.
- **Database & Systems:** Database Management, Role-Based Access Control (RBAC), Server Automation (Cronjobs), System Archival Systems.
- **Core Competencies:** Artificial Intelligence (AI) & Machine Learning, Internet of Things (IoT) Systems, Hardware Troubleshooting, Software Development Life Cycle (SDLC).
- **Soft Skills & Software:** Leadership (Chairperson), Event Planning, Microsoft Office (Word, Excel, PowerPoint).
- **Languages:** Indonesian (Native), English (Professional/Active)